

24/04/2025



Aakash

Medical | IIT-JEE | Foundations

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CODE-A



AIM - 720

(Advanced **INTENSIVE** Mastery for 720)

MM : 720

CST-10

Time : 180 Mins.

Answers

1. (2)	37. (4)	73. (1)	109. (1)	145. (4)
2. (3)	38. (3)	74. (3)	110. (3)	146. (2)
3. (1)	39. (3)	75. (2)	111. (1)	147. (2)
4. (3)	40. (2)	76. (3)	112. (4)	148. (2)
5. (2)	41. (2)	77. (1)	113. (4)	149. (4)
6. (1)	42. (1)	78. (2)	114. (4)	150. (3)
7. (2)	43. (2)	79. (4)	115. (3)	151. (4)
8. (4)	44. (4)	80. (1)	116. (1)	152. (3)
9. (4)	45. (3)	81. (3)	117. (4)	153. (2)
10. (1)	46. (4)	82. (2)	118. (3)	154. (4)
11. (3)	47. (2)	83. (1)	119. (2)	155. (4)
12. (3)	48. (1)	84. (4)	120. (3)	156. (2)
13. (1)	49. (4)	85. (1)	121. (3)	157. (1)
14. (3)	50. (3)	86. (1)	122. (2)	158. (3)
15. (2)	51. (3)	87. (2)	123. (2)	159. (2)
16. (4)	52. (4)	88. (3)	124. (3)	160. (3)
17. (3)	53. (1)	89. (2)	125. (3)	161. (2)
18. (2)	54. (3)	90. (2)	126. (2)	162. (2)
19. (2)	55. (3)	91. (2)	127. (2)	163. (2)
20. (1)	56. (3)	92. (3)	128. (2)	164. (1)
21. (2)	57. (3)	93. (1)	129. (2)	165. (2)
22. (2)	58. (2)	94. (1)	130. (2)	166. (4)
23. (2)	59. (3)	95. (4)	131. (2)	167. (3)
24. (4)	60. (3)	96. (3)	132. (2)	168. (4)
25. (2)	61. (1)	97. (3)	133. (2)	169. (4)
26. (4)	62. (4)	98. (4)	134. (4)	170. (2)
27. (2)	63. (1)	99. (2)	135. (4)	171. (4)
28. (1)	64. (4)	100. (3)	136. (4)	172. (4)
29. (4)	65. (4)	101. (2)	137. (2)	173. (1)
30. (4)	66. (2)	102. (3)	138. (1)	174. (1)
31. (1)	67. (1)	103. (3)	139. (2)	175. (4)
32. (4)	68. (3)	104. (4)	140. (1)	176. (3)
33. (3)	69. (1)	105. (4)	141. (3)	177. (4)
34. (2)	70. (2)	106. (1)	142. (3)	178. (3)
35. (4)	71. (1)	107. (3)	143. (1)	179. (3)
36. (2)	72. (2)	108. (4)	144. (4)	180. (3)

Hints and Solutions

PHYSICS

(1) Answer : (2)

Solution:

With increases in temperature, elasticity of wire decreases.

(2) Answer : (3)

Solution:

$$|\text{Acceleration}| = \omega^2 x$$

$$a_{\text{mean}} = 0 \quad (x = 0)$$

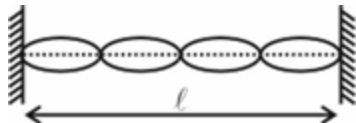
$$v = \omega \sqrt{A^2 - x^2}$$

 At extreme position $x = A$, $v = 0$

$$\text{Total energy} = \frac{1}{2} m \omega^2 A^2 = \text{constant}$$

(3) Answer : (1)

Solution:



$$\frac{4\lambda}{2} = l$$

$$2\lambda = 10$$

$$\lambda = 5 \text{ m}$$

$$\text{velocity} = \lambda v$$

$$\frac{20 \text{ m/s}}{5 \text{ m}} = 4 \text{ Hz} = \nu$$

(4) Answer : (3)

Solution:

For first minima

$$a \sin \theta = \lambda$$

$$a \sin 30^\circ = \lambda$$

$$a = 2\lambda$$

$$= 2 \times 4000 \times 10^{-10}$$

$$= 8 \times 10^{-7}$$

$$= 0.8 \mu\text{m}$$

(5) Answer : (2)

Solution:

$$I_d = C \frac{dV}{dt}$$

$$= C_0 \frac{d}{dt} (V_0 \sin \omega t)$$

$$= C_0 V_0 \omega \cos \omega t$$

(6) Answer : (1)

Solution:

Diamagnetism is universal property. Magnetic susceptibility is constant and negative. Superconductors are diamagnetic.

(7) Answer : (2)

Solution:

$$\vec{B} = \vec{B}_1 + \vec{B}_2$$

$$\vec{B}_1 = \frac{\mu_0 I}{2(2R)} \odot$$

$$\vec{B}_2 = \frac{\mu_0 I}{2(4R)} \otimes$$

$$B = \frac{\mu_0 I}{4R} - \frac{\mu_0 I}{8R}$$

$$= \frac{\mu_0 I}{8R}$$

(8) Answer : (4)

Solution:

$$T = \frac{2\pi m}{qB}$$

(9) Answer : (4)

Solution:

$$\text{Power factor} = \frac{R}{\sqrt{R^2 + \left(\frac{1}{\omega_0 C}\right)^2}}$$

$$\left(\frac{1}{3}\right)^2 = \frac{R^2}{R^2 + \left(\frac{1}{\omega_0 C}\right)^2}$$

$$\left(\frac{1}{\omega C}\right)^2 = 8R^2$$

$$2\sqrt{2}R = \frac{1}{\omega C}$$

$$C = \frac{1}{2\sqrt{2}\pi \times 10^4}$$

$$= \frac{25\sqrt{2}}{\pi} \mu\text{F}$$

(10) Answer : (1)

Solution:

Core of transformer is made up of soft iron.

(11) Answer : (3)

Solution:

$$|\varepsilon| = L \frac{dI}{dt}$$

$$\varepsilon = L \frac{d}{dt}(2t + 3t^2)$$

$$= L(2 + 6t)$$

$$= 4(2 + 6 \times 2) \times 10^{-3}$$

$$= 56 \text{ mV}$$

(12) Answer : (3)

Solution:

$$\varepsilon = -\frac{d\phi}{dt}$$

(13) Answer : (1)

Solution:

$$\text{Torque about } O = F_1 R - F_2 \frac{R}{3} = R \left(F_1 - \frac{F_2}{3} \right)$$

$$= R \left(30 - \frac{60}{3} \right) = 10R$$

$$\Rightarrow \tau = 10R = \frac{MR^2}{2} \alpha$$

$$\Rightarrow \alpha = \frac{20}{MR} = \frac{20}{30 \times \frac{1}{5}} = \frac{10}{3} \text{ rad/s}^2$$

(14) Answer : (3)

Solution:

In a couple, total external force is zero because both forces are equal in magnitude but opposite in direction. The total torque on a system is independent of the origin if the total external force is zero.

(15) Answer : (2)

Solution:

Focal length of the plano-convex lens = f

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R} - 0 \right)$$

$$\Rightarrow f = \frac{R}{\mu - 1} = \frac{20}{1.4 - 1} = \frac{20}{0.4} = 50 \text{ cm}$$

Focal length of the silvered lens:

$$\frac{-1}{f} = \frac{2}{f_{\text{lens}}} - \frac{1}{f_m} \Rightarrow \frac{-1}{f} = \frac{2}{50}$$

$$f = -25 \text{ cm}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{-50} = \frac{1}{-25}$$

$$\frac{1}{v} = \frac{-1}{25} + \frac{1}{50} = \frac{-1}{50} \Rightarrow v = -50 \text{ cm}$$

(16) Answer : (4)

Solution:

In $\triangle AYX$,

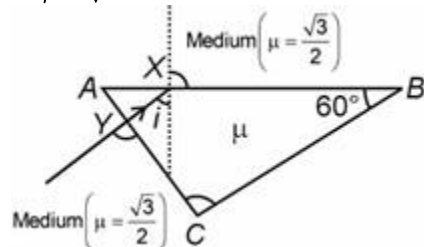
$$30^\circ + 90^\circ + 90^\circ - i = 180^\circ$$

$$\Rightarrow i = 30^\circ$$

$$\text{From Snell's law, } \mu \sin i = \frac{\sqrt{3}}{2} \sin 90^\circ$$

$$\Rightarrow \mu \times \frac{1}{2} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \mu = \sqrt{3}$$



(17) Answer : (3)

Solution:

Work done $U_f - U_i$

$$= -\frac{GMm}{R+h} - \left(-\frac{GMm}{R}\right) = GMm \left(\frac{1}{R} - \frac{1}{5R}\right)$$

$$= \frac{4}{5} \frac{GMm}{R} = \frac{4}{5} \frac{mgR^2}{R} = \frac{4}{5} mgR \quad \left(\text{Where } g = \frac{GM}{R^2}\right)$$

(18) Answer : (2)

Solution:

Total energy of planet will remain conserved as it moves along the elliptical orbit.

Total energy at A = Total energy at C = $K_C + V_C$

(19) Answer : (2)

Solution:

Number of spectral lines emitted during transition from state n is $N = \frac{n(n-1)}{2}$

$$= \frac{5(4)}{2}$$

$$= 10$$

(20) Answer : (1)

Solution:

Output C is 1 (high) when both inputs A and B are 1 (high)

Therefore, AND gate

(21) Answer : (2)

Solution:

• Nucleus force may be attractive or repulsive depending on the distance between nucleons. Force is attractive for distances larger than 0.8 fm and repulsive if they are separated by distances less than 0.8 fm.

• Nuclear force is independent of electric charge.

(22) Answer : (2)

Solution:

Since zero of the vernier scale lies towards left side of the zero mark of main scale

$\therefore$ Negative zero error

(23) Answer : (2)

Solution:

$$i_L = \frac{16}{1 \times 10^3}$$

$$= 16 \text{ mA}$$

(24) Answer : (4)

Solution:

Surface area $A = 6a^2$

$$= 6 \times (2)^2$$

$$= 24 \text{ cm}^2$$

$$\frac{\Delta A}{A} = \frac{2\Delta a}{a}$$

$$\Delta A = 2A \frac{\Delta a}{a}$$

$$= 2 \times 24 \times \frac{0.01}{2.00}$$

$$= 0.24$$

$$\therefore A = (24 \pm 0.24) \text{ cm}^2$$

(25) **Answer :** (2)

Solution:

$$F = qvB$$

$$B = \frac{F}{qv} = \frac{F}{q \frac{d}{t}} = \frac{F}{Id} = \frac{T}{I} = \frac{\text{Surface tension}}{\text{Current}}$$

(26) **Answer :** (4)

Solution:

Displacement = Change in position

As athlete is at initial position again after running 500 m

$$|\Delta \vec{x}| = 0$$

(27) **Answer :** (2)

Solution:

$$t \propto u$$

$$\frac{t_1}{t_2} = \frac{u_1}{u_2}$$

$$\frac{5}{t_2} = \frac{1}{2}$$

$$\Rightarrow t_2 = 10 \text{ s}$$

(28) **Answer :** (1)

Solution:

Impulse = change in momentum = $m[v - u]$

$$= m \left[\frac{3}{2} - 0 \right]$$

$$= 4 \times \frac{3}{2} = 6 \text{ N s}$$

(29) **Answer :** (4)

Solution:

Centripetal acceleration is responsible for a particle to move in curved path therefore component of acceleration has to be perpendicular to $\vec{v}$

(30) **Answer :** (4)

Solution:

$$m = 2 \text{ kg}$$

$$x = 4t^2 + 4$$

$$v = \frac{dx}{dt} = 8t$$

W.D. = change in K.E.

$$= \frac{1}{2} m (v^2 - u^2)$$

$$= \frac{1}{2} \times 2 (8^2 - 0^2) = 64 \text{ J}$$

(31) **Answer :** (1)

Solution:

When non-conservative forces are involved energy is not conserved rather dissipate.

In elastic collision, kinetic energy is conserve.

(32) **Answer :** (4)

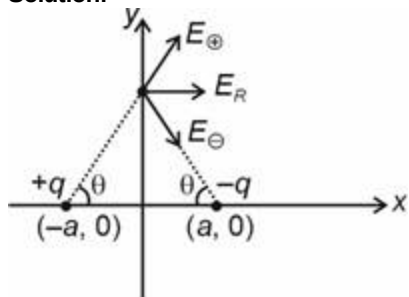
Solution:

Initial uncharged capacitor acts like a short circuit allowing current to flow while fully charged capacitor acts like an open circuit and no further current will flow.

When dielectric is inserted while battery remain connected, then capacitance increases and charge also increases. When dielectric is inserted after removing battery, then capacitance increases but voltage decreases.

(33) Answer : (3)

Solution:



Net electric field on y-axis is along +x axis due to the given dipole system.

(34) Answer : (2)

Solution:

$$F = \frac{kq_1q_2}{r^2}$$

$$12 = \frac{K(2)(2)}{r^2} \dots (i)$$

$$q'_1 = -2 \text{ C}, \quad q'_2 = -6 \text{ C}$$

$$F' = \frac{K(-2)(-6)}{r^2} \dots (ii)$$

$$\frac{12}{F'} = \frac{K(2)(2)}{K(2)(6)}$$

$\Rightarrow F' = 36 \text{ N}$, repulsive because both charges have same sign.

(35) Answer : (4)

Solution:

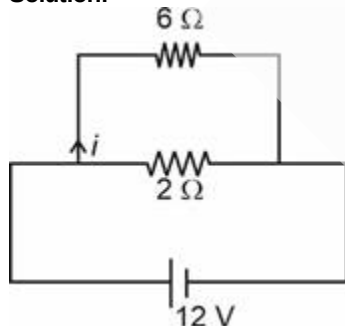
Electric field due to an infinitely wire is

$$E = \frac{2K\lambda}{r}, \text{ where } \lambda \text{ is uniform linear charge density}$$

$$\therefore E \propto \frac{1}{r}$$

(36) Answer : (2)

Solution:



$$V = iR$$

$$12 = i(6)$$

$$\Rightarrow i = 2 \text{ A}$$

(37) Answer : (4)

Solution:

$$R = \frac{\rho l}{A} = \frac{\rho V}{A^2}$$

Where ρ is resistivity that is constant for wire and V is volume and it will also remain constant.

$$R' = \frac{\rho V}{(2A)^2} = \frac{\rho V}{4A^2}$$

$$\Rightarrow R' = \frac{R}{4}$$

$$\% \text{ change in resistance} = \frac{R' - R}{R} \times 100$$

$$= \frac{\frac{R}{4} - R}{R} \times 100 = \frac{3}{4} \times 100 = 75\%$$

(38) Answer : (3)

Solution:

In equilibrium

$$F_B = mg$$

$$V_{in} \rho_w g = V \sigma g$$

$$\frac{V_{in}}{V} = \frac{\sigma}{\rho_w} = \frac{1}{3}$$

$$\therefore \frac{V_{out}}{V} = 1 - \frac{1}{3} = \frac{2}{3}$$

(39) Answer : (3)

Solution:

Let θ be temperature of interface, then in series heat current must be same.

$$\text{i.e. } \frac{\Delta\theta_1}{R_1} = \frac{\Delta\theta_2}{R_2}, R = \frac{1}{KA}$$

$$\Rightarrow \frac{2KA}{1}(100 - \theta) = \frac{KA}{1}(\theta - 40)$$

$$\Rightarrow 200 - 20 = \theta - 40$$

$$240 = 3\theta$$

$$\Rightarrow \boxed{\theta = 80^\circ \text{C}}$$

(40) Answer : (2)

Solution:

Using $TV^{\gamma-1} = \text{constant}$

$$\Rightarrow 273(V)^{\frac{4}{3}-1} = T\left(\frac{V}{8}\right)^{\frac{4}{3}-1}$$

$$\Rightarrow 273 [8]^{1/3} = T$$

$$T = 546 \text{ K}$$

$$\Delta T = 273 \text{ K}$$

(41) Answer : (2)

Solution:

$$W_{ABCD} = W_{AB} + W_{BC} + W_{CD}$$

$$= P_0 V_0 + 0 - 2P_0 V_0$$

$$= -P_0 V_0$$

(42) Answer : (1)

Solution:

$$\gamma_{\text{mix}} = 1 + \frac{2}{f_{\text{mix}}}$$

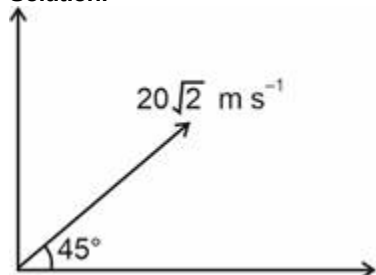
$$f_{\text{mix}} = \frac{n_1 f_1 + n_2 f_2}{n_1 + n_2} = \frac{(1 \times 3) + (2 \times 5)}{1 + 2}$$

$$f_{\text{mix}} = \frac{13}{3}$$

$$\gamma_{\text{mix}} = 1 + \frac{2}{\frac{13}{3}} = \frac{19}{13}$$

(43) Answer : (2)

Solution:



• Maximum height $H = \frac{u^2 \sin^2 \theta}{2g} = \frac{400}{20} = 20 \text{ m}$

• Time of flight $T = \frac{2u \sin \theta}{g} = \frac{40}{10} = 4 \text{ s}$

• Horizontal range

$$R = \frac{u^2 \sin^2 \theta}{g} = \frac{800 \times 1}{10} = 80 \text{ m}$$

• Height of particle at $t = 3 \text{ s}$ is same as that at $t = 1 \text{ s}$

$$h = u_y t - \frac{1}{2} g t^2$$

$$h = 20(1) - \frac{1}{2}(10)(1)^2$$

$$= 15 \text{ m}$$

(44) Answer : (4)

Solution:

De Broglie wavelength $\lambda = \frac{h}{p}$

Since momentum p is same for all particles.

$\therefore \lambda$ is same

(45) Answer : (3)

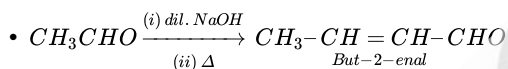
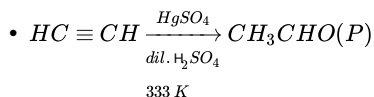
Solution:

Pressure at the bottom of surface is given by $P = hpg$

CHEMISTRY

(46) Answer : (4)

Solution:



• CH_3CHO reduces Fehling's reagent and gives positive iodoform test.

(47) Answer : (2)

Solution:

Insulin is an example of globular protein.

(48) Answer : (1)

Solution:

$$\text{KE} = h\nu - h\nu_0$$

$$\text{KE} = h(\nu - \nu_0)$$

$$= 6.6 \times 10^{-34} (1 \times 10^{16} - 2 \times 10^{15})$$

$$= 6.6 \times 10^{-34} (10 \times 10^5 - 2 \times 10^{15})$$

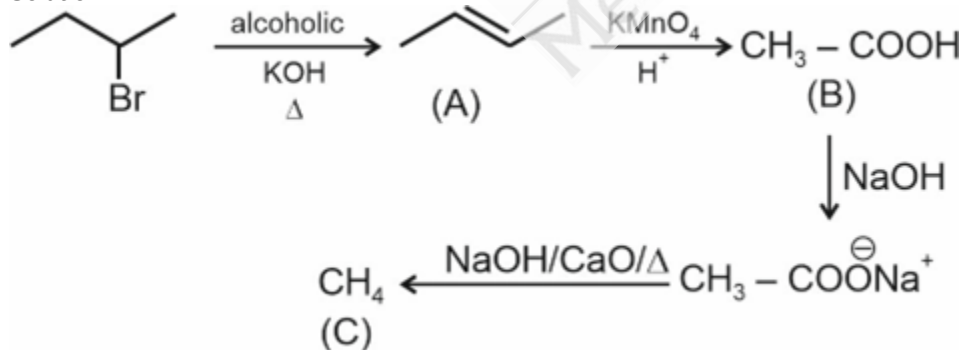
$$= 6.6 \times 10^{-34} \times 8 \times 10^{15}$$

$$= 52.8 \times 10^{-19}$$

$$\text{KE} = 5.28 \times 10^{-18} \text{ J}$$

(49) Answer : (4)

Solution:



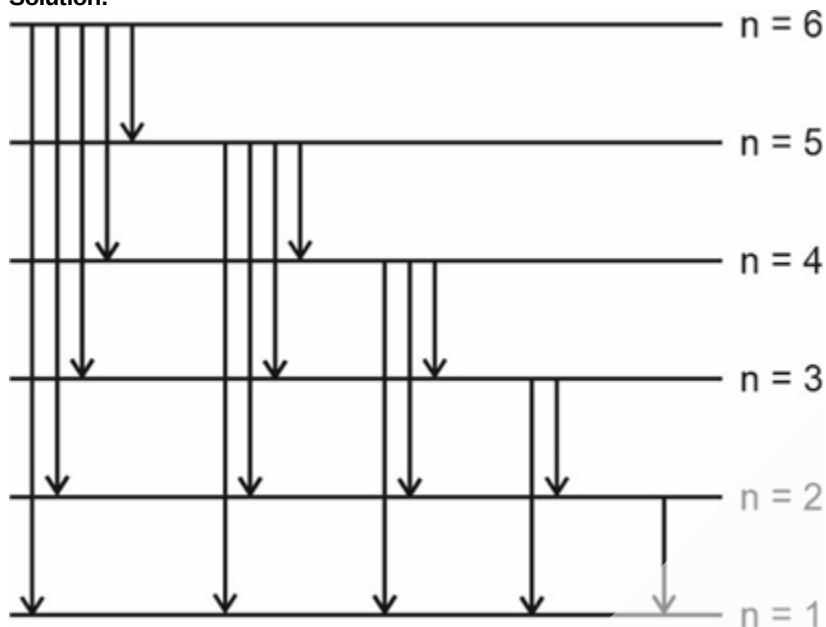
(50) Answer : (3)

Solution:

In DNA molecule, adenine forms hydrogen bond with thymine whereas cytosine forms hydrogen bonds with guanine

(51) Answer : (3)

Solution:

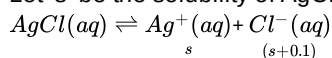


Total number of spectral lines obtained = 15

(52) Answer : (4)

Solution:

Let 's' be the solubility of AgCl.



$$K_{sp} = s(s + 0.1)$$

as $K_{sp} = 1.6 \times 10^{-10}$, is very low hence

$$s \ll 0.1,$$

$$K_{sp} = s(0.1)$$

$$\frac{1.6 \times 10^{-10}}{0.1} = s$$

$$s = 1.6 \times 10^{-9} \text{ M}$$

(53) Answer : (1)

Solution:

$$pH = 7 + \frac{1}{2} (pK_a + \log C)$$

$$= 7 + \frac{1}{2} (4.2 + \log 0.4)$$

$$= 8.9$$

(54) Answer : (3)

Solution:

$$\text{Number of Faradays passed} = \frac{90 \times 60 \times 4.825}{96500}$$

$$= 0.27$$

$$\text{Mass of copper deposited} = \frac{0.27 \times 63.5}{2}$$

$$= 8.57$$

$$= 8.6 \text{ g}$$

(55) Answer : (3)

Solution:

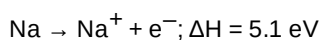
$$\Lambda_m = \frac{1000 \times k}{C}$$

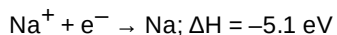
$$\Lambda_m = \frac{0.0248 \times 1000}{0.1}$$

$$\Lambda_m = 248 \text{ S cm}^2 \text{ mol}^{-1}$$

(56) Answer : (3)

Solution:

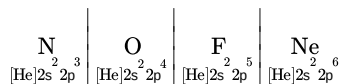




So, electron gain enthalpy of Na^+ is -5.1 eV

(57) Answer : (3)

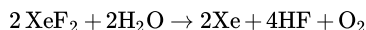
Solution:



Ionisation enthalpy order: $\text{Ne} > \text{F} > \text{N} > \text{O}$

(58) Answer : (2)

Solution:



(59) Answer : (3)

Solution:

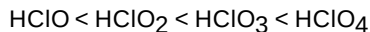
Due to absence of d-orbital in valence shell of nitrogen atom, nitrogen can not form 5 bond. Therefore NCl_5 does not exist.

(60) Answer : (3)

Solution:

As the oxidation state of Halogen increases acidic strength of oxyacid increases.

Therefore correct order of acidic strength is



(61) Answer : (1)

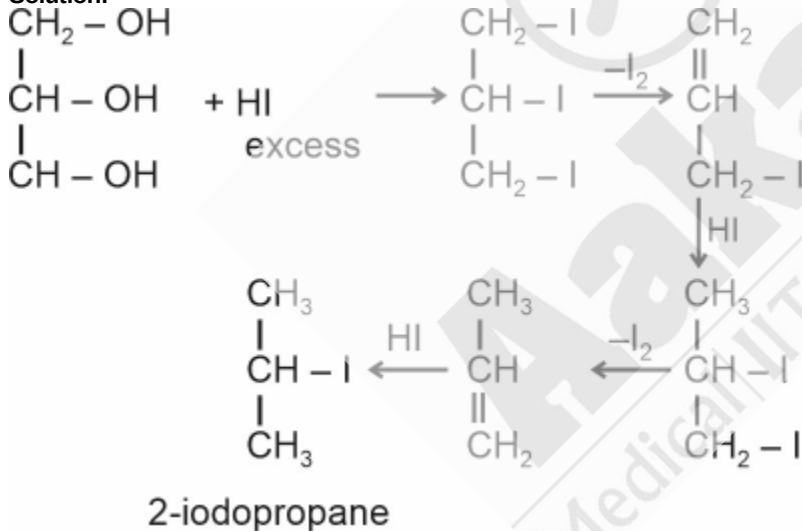
Solution:

Compound which is more acidic than carbonic acid will evolve CO_2 on reaction with NaHCO_3 .

2,4,6-trinitrophenol and benzoic acid are stronger acid than carbonic acid.

(62) Answer : (4)

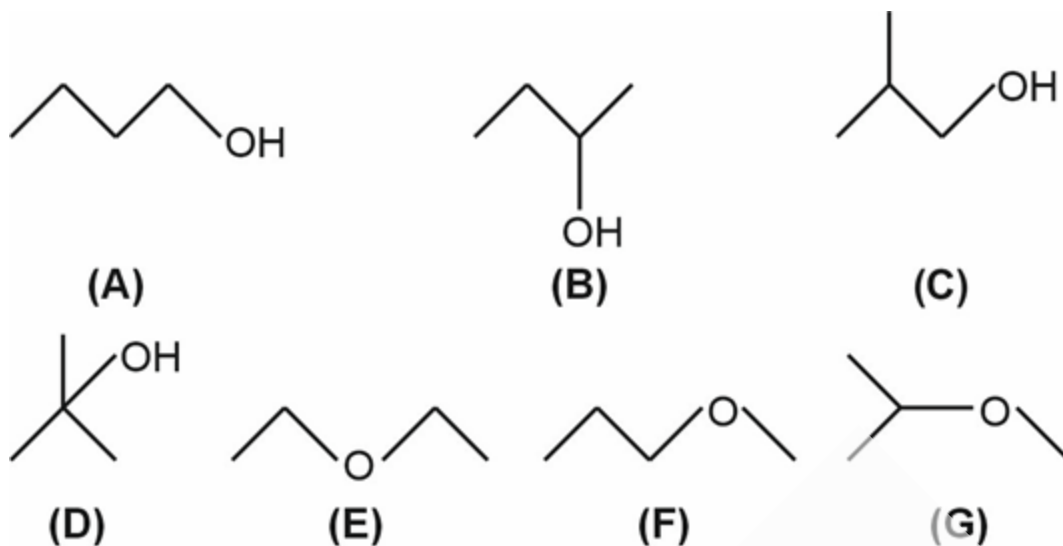
Solution:



(63) Answer : (1)

Solution:

Total 8 isomers are possible for molecule formula $\text{C}_4\text{H}_{10}\text{O}$



• (B) exists as a pair of enantiomers.

(64) Answer : (4)

Solution:

- (i) Resonating structures are hypothetical and do not represent any real molecule.
- (ii) The energy of actual structure of the molecule (resonance hybrid) is lower than that of any canonical structures.
- (iii) The resonance structures have the same position of nuclei and the same number of unpaired electrons.

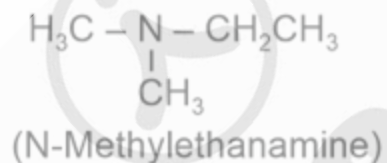
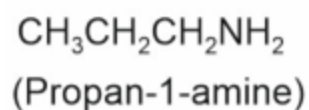
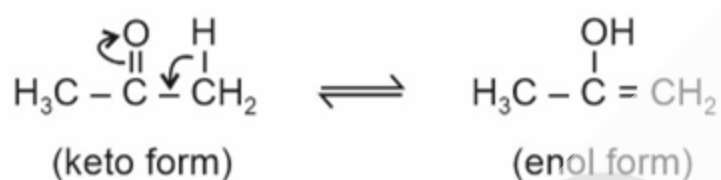
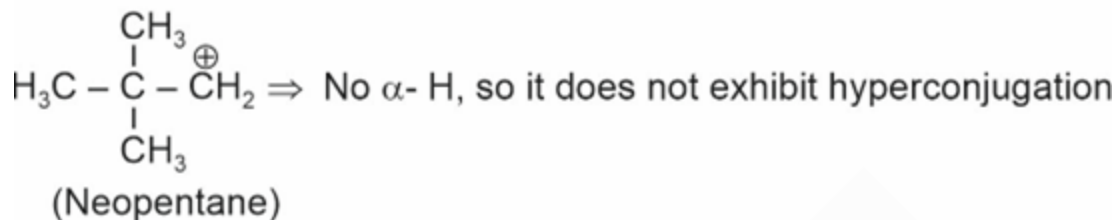
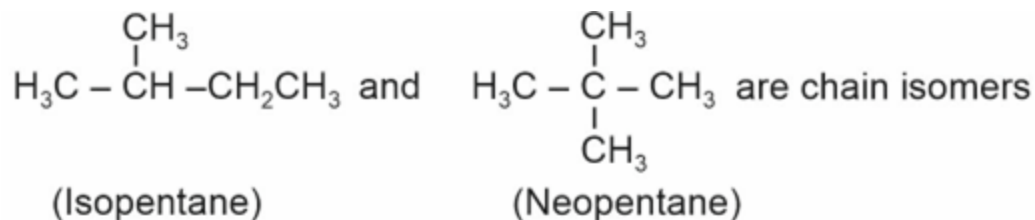
(65) Answer : (4)

Solution:

- | | |
|-----------------|---|
| Pyridine | (i) Heterocyclic aromatic compound. |
| Tetrahydrofuran | (ii) Heterocyclic non aromatic compound |
| Tropone | (iii) Non-benzenoid aromatic compound |
| Naphthalene | (iv) Benzenoid aromatic compound |

(66) Answer : (2)

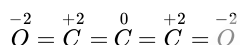
Solution:



$\Rightarrow$ functional group isomers

(67) Answer : (1)

Solution:



(68) Answer : (3)

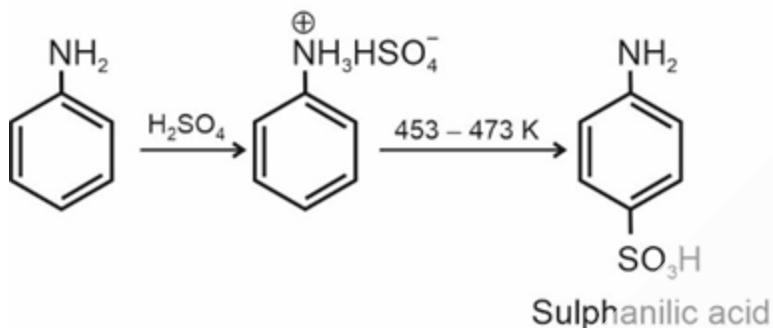
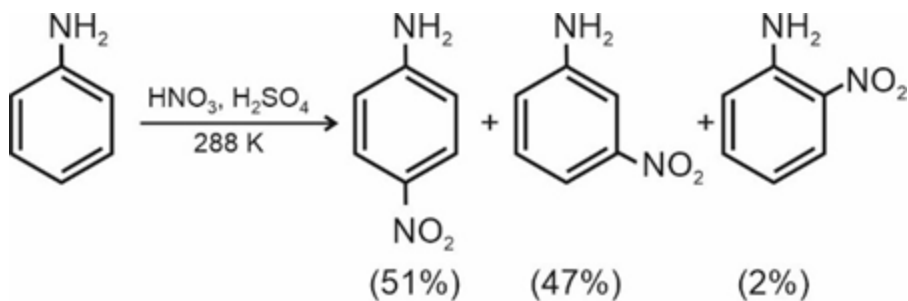
Solution:

Water gas – CO + H₂

Producer gas – CO + N₂

(69) Answer : (1)

Solution:



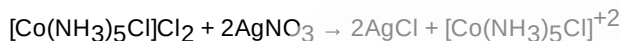
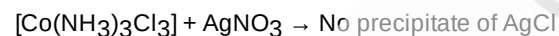
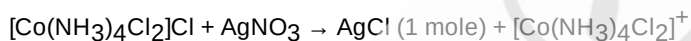
(70) Answer : (2)

Solution:

Secondary amines react with Hinsberg's reagent to form N, N-Dialkylbenzene sulphonamide, which is insoluble in alkali.

(71) Answer : (1)

Solution:



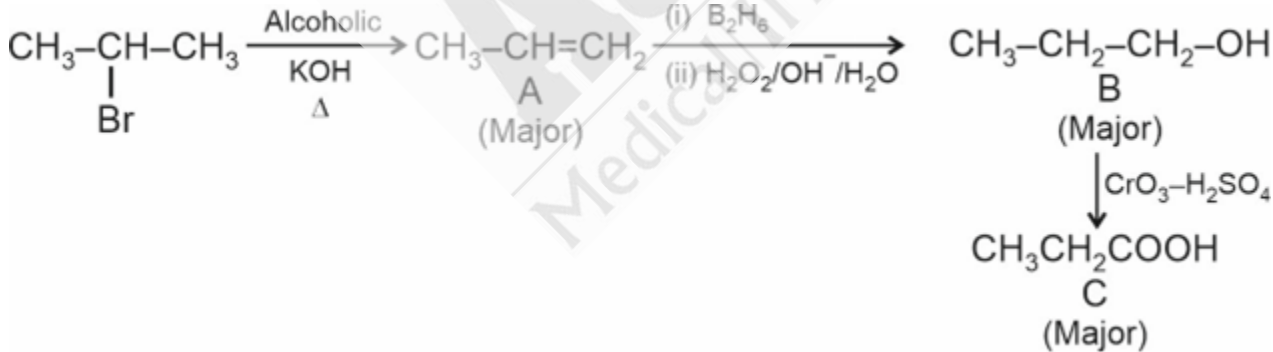
(72) Answer : (2)

Solution:

More is the negative charge on metal carbonyl, more is back donation, weaker is C – O bond and longer is C – O bond length.

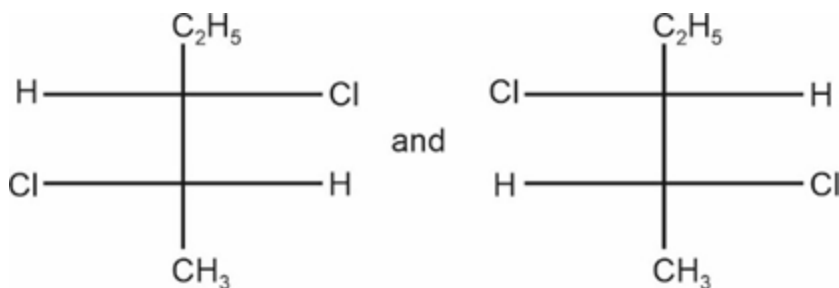
(73) Answer : (1)

Solution:



(74) Answer : (3)

Solution:

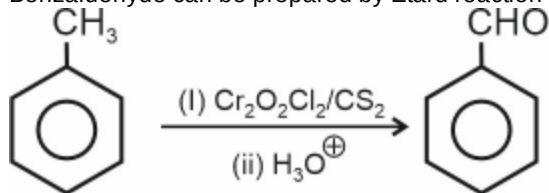


are enantiomers to each other because they are mirror images to each other but do not superimpose.

(75) Answer : (2)

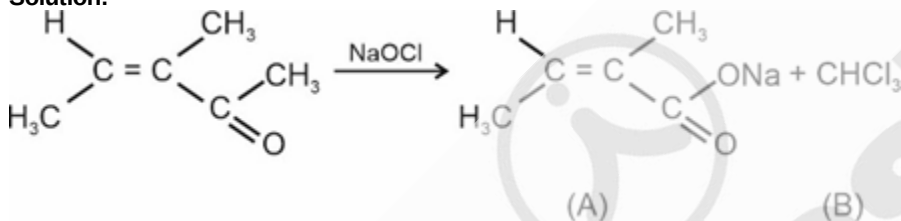
Solution:

Benzaldehyde can be prepared by Etard reaction



(76) Answer : (3)

Solution:



(77) Answer : (1)

Solution:

Slow step is rate determining step

$$r = k[A_3][A]$$

$$K_C = \frac{[A_2][A]}{[A_3]}$$

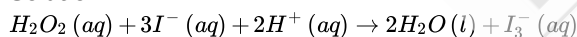
$$[A] = \frac{K_C[A_3]}{[A_2]}$$

$$r = KK_C \frac{[A_3]}{[A_2]} [A_3]$$

$$r = k[A_3]^2[A_2]^{-1}$$

(78) Answer : (2)

Solution:



$$\frac{-d[H_2O_2]}{dt} = \frac{-1}{3} \frac{d[I^-]}{dt} = \frac{-1}{2} \frac{d[H^+]}{dt} = \frac{1}{2} \frac{d[H_2O]}{dt} = \frac{d[I_3^-]}{dt}$$

$$\frac{d[I_3^-]}{dt} = \frac{1}{3}x$$

(79) Answer : (4)

Solution:

Group-I – dil. HCl

Group-II – H_2S + dil. HCl

Group-III – $NH_4OH + NH_4Cl$

Group-IV – $H_2S + NH_4OH$

(80) Answer : (1)

Solution:

Substances in their standard form have zero value of enthalpy of formation.

(81) Answer : (3)

Solution:

$$\Delta T_f = iK_f m$$

$$\Delta T_f \propto m$$

$$\Delta T_f (\text{glucose}) = 1 \times K_f \times 0.1 = 0.1 K_f$$

$$\Delta T_f (\text{NaCl}) = 2 \times K_f \times 0.1 = 0.2 K_f$$

(82) Answer : (2)

Solution:

Let the molar mass of solute be x.

$$\frac{P^\circ - P_s}{P^\circ} = \frac{n_B}{n_A + n_B}$$

$$\frac{20}{100} = \frac{\frac{10}{x}}{\frac{10}{x} + 1}$$

$$\frac{2}{10} \left(\frac{10}{x} + 1 \right) = \frac{10}{x}$$

$$x = 40$$

(83) Answer : (1)

Solution:

In CO_3^{2-} only $p\pi-p\pi$ bond is present

(84) Answer : (4)

Solution:

Bond order of O_2 is 2

whereas O_2^+ is 2.5

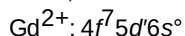
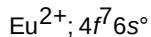
(85) Answer : (1)

Solution:

Due to high mutual repulsion electron present in antibonding molecular orbital destabilize the molecule.

(86) Answer : (1)

Solution:



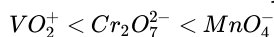
Both are paramagnetic in nature

• $\text{Nd}(\text{OH})_3$ is more basic in nature than $\text{Gd}(\text{OH})_3$

(87) Answer : (2)

Solution:

Due to increased stability of the lower species to which the molecules are reduce, the order of oxidising power is



(88) Answer : (3)

Solution:

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$= R_H \left(\frac{1}{2^2} - \frac{1}{5^2} \right)$$

$$\frac{1}{\lambda} = R_H \left(\frac{25-4}{100} \right) = R_H \frac{21}{100}$$

$$\lambda = \frac{100}{21R_H}$$

(89) Answer : (2)

Solution:

Catalyst does not affect the equilibrium composition of a reaction mixture.

(90) Answer : (2)

Solution:

The SI unit for amount of substance is mole.

The SI unit for mass is kg.

BIOLOGY

(91) Answer : (2)

Solution:

Taxonomic categories showing hierarchial arrangement in ascending order is

Triticum → Poaceae → Poales → Monocotyledonae

↓ ↓ ↓ ↓
Genus Family Order Class

(92) Answer : (3)

Solution:

Euglenoids are photosynthetic in presence of sunlight. They have the pigments identical to those present in higher plants. Protozoans are believed to be primitive relatives of animals. They are heterotrophs and live as predators or parasites.

(93) Answer : (1)

Solution:

Sleeping sickness is caused by *Trypanosoma*.
BSE is caused by prions.

(94) Answer : (1)

Solution:

The floral formula of Liliaceae family is



Leaves: mostly basal, alternate, linear, exstipulate with parallel venation. Seeds are endospermous.

(95) Answer : (4)

Solution:

The plumule and radicle are enclosed in sheaths which are called coleoptile and coleorhiza respectively.

(96) Answer : (3)

Solution:

Trifolium belongs to Fabaceae family.

(97) Answer : (3)

Solution:

Puccinia is an example of Basidiomycetes and it is a rust fungus. Conidia are not found in *Puccinia*.

(98) Answer : (4)

Solution:

Fimbriae and gas vacuole are exclusively found in the prokaryotic cells. Whereas flagella, ribosomes and cell wall can be observed in eukaryotic cells also.

(99) Answer : (2)

Solution:

The axoneme usually has nine doublets of radially arranged peripheral microtubules and pair of centrally located microtubules.

(100) Answer : (3)

Solution:

In S phase, there is doubling of the DNA content.

In G₂ phase, DNA content will be 40 pg.

After meiosis I, the amount of DNA will be equally divided into the 2 daughter cells and it will be 20 pg in each.

During Meiosis II, DNA content is further halved. The reduced DNA content will be 10 pg in each daughter cell.

(101) Answer : (2)

Solution:

Attachment of spindle fibres to the kinetochores occurs in the metaphase stage.

Chromatin condensation is an event of prophase.

Exchange of genetic material between homologous chromosomes occurs in meiosis I.

(102) Answer : (3)

Solution:

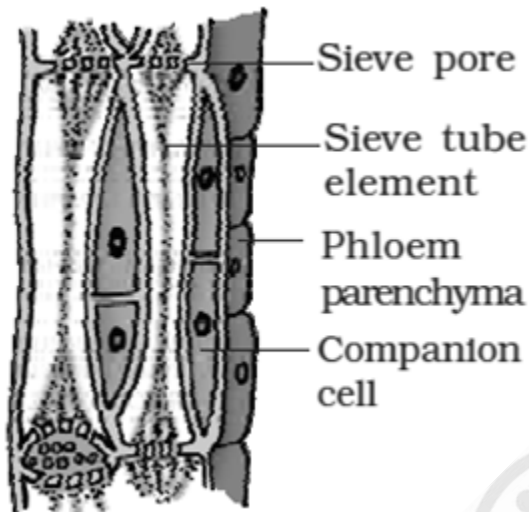
True nucleus is observed in eukaryotes. Amongst the mentioned organisms, only *Volvox* (an alga) is a eukaryote.

(103) Answer : (3)**Solution:**

Kinetochores are not a part of the mitotic apparatus.

(104) Answer : (4)**Solution:**

Parenchyma forms the major component within plant organs and generally has isodiametric cells.

(105) Answer : (4)**Solution:**

Companion cells are helpful in maintaining pressure gradient in sieve tubes. These are absent in gymnosperms and pteridophytes.

Phloem fibres are the only dead element of phloem.

(106) Answer : (1)**Solution:**

'X' is gonadotrophin releasing hormone.

Hypothalamus releases GnRH which stimulates the pituitary to synthesise LH and FSH.

- LH induces ovulation and also maintains the corpus luteum.
- Androgens are responsible for the development of secondary sexual characters in males.
- Progesterone acts on the mammary glands and stimulates the formation of alveoli.

(107) Answer : (3)**Solution:**

The yellowish fluid secreted by the mother during the initial days of lactation has abundant IgA to protect the infant. The foetus also receives some antibodies from the mother through placenta during pregnancy.

Colostrum provides natural passive immunity.

(108) Answer : (4)**Solution:**

Natural cannabinoids are obtained from the inflorescences of plant *Cannabis sativa*.

Opioids like morphine acts as an analgesic that is given to patients post-surgery.

(109) Answer : (1)**Solution:****Benign tumors**

Do not show metastasis

Remain confined to their original location

Malignant tumors

Show metastasis

(110) Answer : (3)**Solution:**

Underproduction of hormones by adrenal cortex alters the carbohydrate metabolism causing acute weakness and fatigue leading to Addison's disease.

Pancreas is a composite gland which acts as both endocrine and exocrine gland.

The endocrine part of pancreas consists of 'Islets of Langerhans'. There are about 1 to 2 million Islets of Langerhans representing only 1 to 2 per cent of the pancreatic tissue.

(111) Answer : (1)**Solution:**

Agarose – Natural polymer extracted from sea weeds

kan^R – Kanamycin resistance encoding gene, but it is not present in pBR322

rop – Codes for the proteins that are involved in the replication of the plasmid

(112) Answer : (4)

Solution:

Gel electrophoresis is commonly used to check the progression of restriction digestion by separating DNA fragments based on size.

Northern blotting is a laboratory technique that detects specific RNA sequences in a sample. It is used to study gene expression.

Probe hybridisation must be followed by autoradiography.

(113) Answer : (4)

Solution:

Archaeogenetics, diagnostics and DNA forensics are just three of the new disciplines that have become possible as a direct consequence of the invention of PCR, enabling molecular biologists to ask questions about human evolution and the impact of environmental change on the biosphere, and to bring their powerful tools to bear in the fight against crime.

(114) Answer : (4)

Solution:

In order to force the bacteria to take up the recombinant plasmid, the bacterial cells must first be made 'competent' to take up DNA. This is done by treating them with a specific concentration of a divalent cation, such as calcium, which increases the efficiency with which DNA enters.

(115) Answer : (3)

Solution:

The ionic gradients across the resting axonal membrane is maintained by the active transport of ions by the sodium-potassium pump which transports 3Na^+ outwards for 2K^+ into the cell.

(116) Answer : (1)

Solution:

Griffith concluded that R strain bacteria had somehow been transformed by heat-killed S-strain bacteria, which had enabled R strain to synthesise a smooth polysaccharide coat and become virulent.

(117) Answer : (4)

Solution:

If both strands of DNA are copied then two RNA molecules if produced simultaneously would be complementary to each other hence would form dsRNA and this would prevent the translation of RNA into protein.

(118) Answer : (3)

Solution:

The human genome contains 3164.7 million bp.

(119) Answer : (2)

Solution:

Bulliform cells – Large, empty and colourless cells found in monocots

Cuticle – Waxy thick layer prevents water loss

Stomata – Composed of dumb-bell shaped guard cells and tiny pore in grasses

Epidermis – Usually single layered and present in most plant parts

(120) Answer : (3)

Solution:

Peptidyl transferase catalyses the formation of peptide bond.

(121) Answer : (3)

Solution:

The process of capping occurs at the 5' end of primary transcript.

(122) Answer : (2)

Solution:

Ethylene promotes seed and bud dormancy and initiates germination in peanut seeds.

(123) Answer : (2)

Solution:

Gibberellins are the terpene derivatives, not the indole compounds.

(124) Answer : (3)

Solution:

Exine of pollen grains exhibits a fascinating array of patterns and designs which is of taxonomic significance.

(125) Answer : (3)

Solution:

The given figure represents the multicarpellary, apocarpous gynoecium in *Michelia*.

(126) Answer : (2)

Solution:

Emasculation refers to the process of removal of anthers and hence it is not required for a staminate flower for artificial hybridisation.

Bisexual flowers, showing protogyny can be used for artificial hybridisation.

(127) Answer : (2)

Solution:

In a monohybrid cross, the phenotypic ratio of the F_2 generation is 3 : 1.

(128) Answer : (2)

Solution:

Recessive traits are expressed only when two recessive alleles are present in diploid species.

(129) Answer : (2)

Solution:

The genes that are present on different chromosomes assort independently.

(130) Answer : (2)

Solution:

Sickle cell anaemia – Defect is caused by transversion of gene controlling β -chain of haemoglobin

α -Thalassemia – Controlled by two closely linked genes HBA1 and HBA2 on chromosome 16

Phenylketonuria – Inborn error of metabolism

Haemophilia – Due to presence of defective form of blood clotting factor

(131) Answer : (2)

Solution:

Both neurons and cardiac muscle cells have the ability to respond to stimuli by generating electrical impulses. This property is called excitability, which is crucial for nerve signal transmission and heart contractions.

(132) Answer : (2)

Solution:

Pst I restriction site is present within the ampicillin resistance gene, thus it undergoes insertional inactivation if the gene of interest is ligated at *Pst* I. After the transformation with pBR322, the *E. coli* cells that have taken up only the plasmid will be $amp^R tet^R$ and will be able to form colonies on an agar medium that contains ampicillin or tetracycline; non-transformants will be $amp^S tet^S$.

(133) Answer : (2)

Solution:

Glandular epithellum are mainly of two types: unicellular, consisting of isolated glandular cells (goblet cells of the alimentary canal), and multicellular, consisting of cluster of cells (salivary gland).

Other mentioned glands are endocrine glands.

(134) Answer : (4)

Solution:

The chemical and physical properties of amino acids are essentially of the amino, carboxyl and the 'R' functional groups.

(135) Answer : (4)

Solution:

In cockroach, the head is connected with the thorax by a short extension of the prothorax known as the neck. Labrum and labium are mouth parts of cockroaches.

(136) Answer : (4)

Solution:

The opening between right atrium and right ventricle is guarded by a valve formed by three muscular flaps or cusps. So, this valve is called tricuspid valve.

(137) Answer : (2)

Solution:

Glycogen is a homopolysaccharide and possesses glucose as its monomeric unit. So, it lacks a disulphide bond.

Aspartic acid (amino acid) is considered as a biomolecule.

(138) Answer : (1)**Solution:**

If the columnar or cuboidal cells bear cilia on their free surface, they are called ciliated epithelium. Their function is to move particles or mucus in a specific direction over the epithelium. They are mainly present in the inner surface of hollow organs like bronchioles and fallopian tubes.

PCT is lined by microvilli. Alveoli contain squamous epithelium.

(139) Answer : (2)**Solution:**

Component	% of the total cellular mass
Water	70-90
Proteins	10-15
Carbohydrates	3
Lipids	2
Nucleic acids	5-7
Ions	1

(140) Answer : (1)**Solution:**

In skeletal muscles, contraction is initiated by a signal sent by the CNS via a motor neuron. Repeated activation of the muscles can lead to the accumulation of lactic acid due to anaerobic breakdown of glycogen in them, causing fatigue.

(141) Answer : (3)**Solution:**

Fishes have a 2-chambered heart with an atrium and a ventricle. Amphibians and reptiles (except crocodile) have 3-chambered heart with 2 atria and a single ventricle, whereas, birds and mammals possess a 4-chambered heart with two atria and two ventricles.

(142) Answer : (3)**Solution:**

Monocytes and neutrophils are phagocytic cells. Eosinophils resist infection and are also associated with allergic reactions. Erythrocytes transport respiratory gases.

(143) Answer : (1)**Solution:**

Parietal and temporal bones are paired cranial bones. Frontal, occipital, sphenoid and ethmoid are unpaired cranial bones. Zygomatic is a paired facial bone.

(144) Answer : (4)**Solution:**

In humans, seven cervical vertebrae are present.

Scapula – 2

Ribs – 12 pairs

Pelvic girdle is formed of 2 coxal bones.

(145) Answer : (4)**Solution:**

Birds and mammals are able to maintain a constant body temperature i.e., they are warm-blooded organisms. *Calotes* is a reptile and it is a cold-blooded organism.

(146) Answer : (2)**Solution:**

The Neanderthal man had a cranial capacity of 1400cc.

(147) Answer : (2)**Solution:**

Branching descent and natural selection are the two key concepts of Darwinian theory of evolution. The essence of Darwinian theory about evolution is natural selection.

Variations are small and directional.

(148) Answer : (2)**Solution:**

The correct chronological order of human evolution is

Ramapithecus → *Australopithecines* → *Homo habilis* → *Homo erectus* → Neanderthal man → Modern *Homo sapiens*

(149) Answer : (4)**Solution:**

Reabsorption is minimum in the ascending limb of loop of Henle. However, this region plays a significant role in the maintenance of high osmolarity of medullary interstitial fluid. The descending limb of Henle's loop is permeable to water but almost impermeable to electrolytes.

(150) Answer : (3)

Solution:

Marsupial rat, tiger cat, banded anteater and sugar glider are Australian marsupials and exhibit adaptive radiation. Eyes of *Octopus* and that of dogs are analogous structures which is a result of convergent evolution. Mouse and lemur are placental mammals.

(151) Answer : (4)

Solution:

The flow of filtrate in the two limbs of Henle's loop is in opposite direction and thus forms a counter current. The flow of blood through the two limbs of vasa recta is also in a counter current pattern. The counter current mechanism helps to maintain a concentration gradient in the medullary interstitium that helps in easy passage of water. Thereby, concentrating the filtrate.

(152) Answer : (3)

Solution:

PCR can detect very low concentration of bacteria or virus by amplification of their nucleic acid.

(153) Answer : (2)

Solution:

In hemichordates, circulatory system is of open type. They have proboscis gland for excretion.

In echinoderms, no excretory structure is present. Echinoderms are characterized by calcareous ossicles.

(154) Answer : (4)

Solution:

RNAi involves silencing of a specific mRNA due to complementary dsRNA molecule that binds to and prevents translation of the mRNA.

(155) Answer : (4)

Solution:

Introduction of the gene isolated from marrow cells producing ADA can be into cells at early embryonic stages can be a permanent cure for ADA deficiency.

(156) Answer : (2)

Solution:

The gymnosperms are plants in which the seeds are not enclosed within fruit wall as the ovules are not enclosed inside the ovaries.

(157) Answer : (1)

Solution:

In pteridophytes, the main plant body is a multicellular sporophyte and differentiated into true root, stem and leaves.

(158) Answer : (3)

Solution:

Streptokinases are produced by the bacterium *Streptococcus* and modified by genetic engineering.

(159) Answer : (2)

Solution:

The density of a population in a given habitat during a given period, fluctuates due to changes in four basic processes (natality, immigration, mortality and emigration).

(160) Answer : (3)

Solution:

In our biosphere, immense diversity (heterogeneity) exists not only at the species level but at all levels of biological organisation ranging from macromolecules within cells to biomes.

(161) Answer : (2)

Solution:

Colombia located near the equator has nearly 1,400 species of birds while New York at 41°N has 105 species.

(162) Answer : (2)

Solution:

Members of brown algae vary in colour from olive green to various shades of brown depending upon the amount of the xanthophyll called fucoxanthin.

(163) Answer : (2)

Solution:

Primary treatment basically involves physical removal of particles-large and small, from the sewage through filtration and sedimentation.

(164) Answer : (1)

Solution:

Protocooperation is an association between individuals of two species, each of which is benefitted by the presence of other but can live equally well without association. e.g., sea anemone attached to the body of hermit crab.

(165) Answer : (2)

Solution:

The primary acceptor of the electron in chloroplast is located towards the outer side of the thylakoid membrane. During Calvin cycle, reduction step involves utilisation of two molecules of ATP and NADPH, per molecule of CO₂ fixed.

(166) Answer : (4)

Solution:

FADH₂ is not a product of light reaction of photosynthesis.

(167) Answer : (3)

Solution:

Oxidative decarboxylation reaction in mitochondria is catalyzed by pyruvate dehydrogenase.

(168) Answer : (4)

Solution:

Earthworms break down detritus into smaller particles called fragmentation.

Humus is further degraded to release inorganic nutrients.

Water soluble inorganic nutrient go into the soil and get precipitated by a process called leaching

DFC starts with dead organisms.

The rate of decomposition is controlled by chemical composition of detritus and climatic factors.

(169) Answer : (4)

Solution:

In the given steps of the process of glycolysis, labels are as follows:

A – Phosphofructokinase

B – NADH + H⁺

C – ATP

X – Glyceraldehyde-3- phosphate

Y – Dihydroxy acetone phosphate

Z – 1,3-bisphosphoglyceric acid

(170) Answer : (2)

Solution:

Primary production is defined as the amount of biomass or organic matter produced per unit area over a period of time by plants during photosynthesis.

(171) Answer : (4)

Solution:

Fasciola is an endoparasite. It contains only suckers not hooks.

(172) Answer : (4)

Solution:

The seminal plasma along with the sperms constitute the semen.

(173) Answer : (1)

Solution:

Since one secondary spermatocyte produces two sperms so, 100 million secondary spermatocytes will produce 200 million spermatozoa.

During oogenesis, each primary oocyte produces only one secondary oocyte. So, 200 million secondary oocytes will be produced by 200 million primary oocytes.

(174) Answer : (1)

Solution:

In human females, oogenesis starts during embryonic development. Acrosome is filled with enzymes. A fluid-filled cavity called antrum is present in the tertiary follicle. First polar body is very small in comparison to secondary oocyte.

(175) Answer : (4)

Solution:

Both frogs and cockroaches lay eggs. In cockroaches, development takes place through a number of nymphal stages. In frogs, development takes place through a larval stage called tadpole.

(176) Answer : (3)

Solution:

Volume of air inspired or expired during a normal respiration is called tidal volume. It is approx. 500 mL *i.e.*, a healthy man can inspire or expire approximately 6000 to 8000 mL of air per minute. This is known as minute volume of respiration which can be represented by Tidal volume X Breathing rate.

(177) Answer : (4)

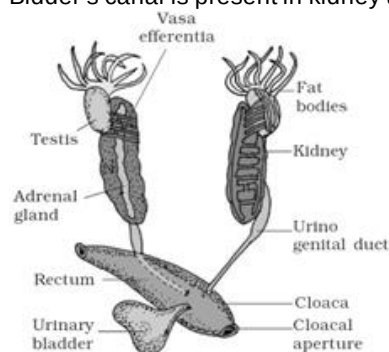
Solution:

The anatomical setup of lungs in thorax is such that any change in the volume of the thoracic cavity will be reflected in the lung (pulmonary) cavity. Such an arrangement is essential for breathing as we cannot directly alter the pulmonary volume. During normal expiration, intra-pulmonary pressure increases slightly above the atmospheric pressure.

(178) Answer : (3)

Solution:

Bidder's canal is present in kidney and is responsible for passage of sperms.



(179) Answer : (3)

Solution:

Pills have to be taken daily for a period of 21 days starting preferably within the first five days of menstrual cycle. After a gap of 7 days during which menstruation occurs, it has to be repeated in the same pattern till the female desires to prevent conception.

They thicken the cervical mucus.

(180) Answer : (3)

Solution:

In vasectomy, a part of vas deferens is cut and tied to block the transport of sperms.